

Engineering Trustworthy AI: A Developer Guide for Empirical Risk Minimization

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Abstract—AI systems are increasingly used for critical decisions that transcend all important fields of private and public life. These systems often use empirical risk minimization (ERM) to train powerful prediction models such as deep neural networks. The output of the predictive model runs the risk of unintentional bias, opacity, and other adverse effects. To ensure the safety of these systems, it is vital to consider these risks already in the design stage of the model. The EU acknowledged the potential sensitivity of the predictions/decisions made about persons, which led to the formulation of the Ethics Guidelines for Trustworthy AI laying down seven key requirements for trustworthy AI. So far, the design of ERM-based methods prioritises accuracy over trustworthiness. This article discusses how key requirements for trustworthy AI can be translated into design choices for ERM components. We map the design space of ML systems to the core objectives of trustworthy AI: fairness, privacy, robustness, and explainability. Our theory is instrumental in the design of trustworthy ML systems that minimize privacy leakage and are robust against (intentional) perturbations during their operation, such as disseminating fake news. The operation of trustworthy ML systems should also be transparent or explainable to its users. Finally, ML systems must be fair and not discriminate against specific user groups. There is an urgent need for a more holistic approach to ML that includes key requirements for trustworthy AI.

Impact Statement—AI systems are used to make predictions and decisions about people in many fields of public and private life, ranging from law enforcement to advertisement. It is vital that AI systems follow minimum requirements for robustness, safety, transparency, privacy and governance, human agency and oversight, and nondiscrimination, and do not cause harm to the persons they make decisions about. As the context in which AI systems are used changes and there are many different AI applications, it is important to have a wide variety of methods available to developers to make sensible design choices for trustworthy AI. Our contribution consists in proposing and showcasing different design choices for ERM components to fulfill the requirements of trustworthy AI. This is an important step to make AI systems safe and socially beneficial and to meet

legislative requirements, such as the General Data Protection Regulation and the AI Act.

Index Terms—AI ethics, empirical risk minimization (ERM), responsible AI design, trustworthy AI.

I. INTRODUCTION

ARTIFICIAL intelligence (AI) has become an integral part of daily life, influencing areas such as job search, housing, and finding new relationships [1], [2]. AI systems often employ machine learning (ML) to train personalized models for users. These trained models provide tailored predictions about interests such as job offers, dating, news, and music videos [3]. Personalized predictions are widely used and too many a matter of convenience. Importantly, the use of personalization in areas such as healthcare, providing suggestions for diagnosis and treatment, can significantly improve people's lives [4].

A. Anecdotal AI Trust Concerns

Despite the usefulness of ML applications, there is growing evidence of their potentially harmful effects.

- 1) *Impact on Democratic Processes*: Democratic processes are the very foundation of working democracies. In recent years, there have been several concerns about undue influences on elections and overall political opinions, polarisation, and radicalisation through social media platforms [5]. Social media platforms have become the center of discussion for the spread of (political) misinformation, propaganda, and hate speech. In particular recommender systems used on these platforms are problematic in this regard as these systems can select (or suppress) information presented to a user [6]. These recommender systems can be exploited to promote sensationalist and divisive content, which may, in turn, intensify polarisation and fragment the public sphere into isolated filter bubbles [7], [8]. Importantly, also public actors have deployed worrisome AI systems, such as COMPAS, a recidivism prediction algorithm used in the US until it was found that it disproportionally predicted African American and female defendants as higher risk than white males [9], [10]. Similar attempts were made in the EU when the police in the Netherlands deployed an algorithmic system called “ProKid” to predict whether juveniles would commit crimes in the future [11]. This kind of preemptive policing stands in clear contrast to the presumption of

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innocence [12] and additionally might even harm the child in its development as the clear focus of police on individuals and preemptive measures could become a self-fulfilling prophecy.

- 2) *Autopilot Crashes*: AI-based vehicle control has been associated with several notable accidents. In some cases, the system failed to detect a specific type of obstacle (such as emergency vehicles) or misinterpreted road conditions and traffic signs [13], [14]. AI-based autopilots can also reduce driver engagement and, in turn, awareness of dangerous situations that require human intervention [15]. This case illustrates the importance of requiring AI systems to be transparent about their operation and limitations [16].
- 3) *Uighur minority in China*: Facial recognition has been used to identify members of the Muslim minority group of Uighurs [17], [18], [19]. The use of facial recognition technology to target a particular ethnic group highlights fundamental concerns about the harmful effects or misuse of AI systems. Trustworthy AI must adhere to ethical principles and respect human rights, including the right to privacy and well-being, at both the individual and societal levels.

B. AI Regulation

Due to the omnipresence of AI, the regulation and management of AI development and risks in all possible application domains is a major challenge. The use of AI has the potential to be harmful far beyond individual levels and can potentially put entire democracies into jeopardy. The EU reacted accordingly with its own AI strategy, aimed at guiding AI development and ultimately passed the AI Act. In the course of this strategy, the EU also published the nonbinding Guidelines on Trustworthy AI, which are examined in the following and lay down seven key requirements (KR1–KR7) for trustworthy AI [20]. It is important to note that while we base our proposals on the European Guidelines, other organizations and countries have proposed similar ethical standards, such as Australia [21] and the organisation for economic co-operation and development (OECD) [22]. Additionally, an analysis of 84 ethics guidelines from around the world revealed several recurring themes that are also reflected in the European Guidelines, such as transparency, justice, fairness and equity, nonmaleficence, responsibility and accountability, and privacy [23]. Therefore, our suggestions can also be evaluated for their relevance to these guidelines beyond the EU.

C. Connecting ERM to Trustworthy AI Requirements

Empirical risk minimization (ERM) is the foundational optimization principle underlying most modern ML systems [24], [25]. The idea of ERM is intuitive: train a ML model by minimizing the average error (or loss) incurred by its predictions on a training set. Design choices within ERM, specifically the construction of the dataset, the selection of the model architecture, and the error measure (or loss), directly influence the ability of AI systems to meet key requirements for trustworthiness (see

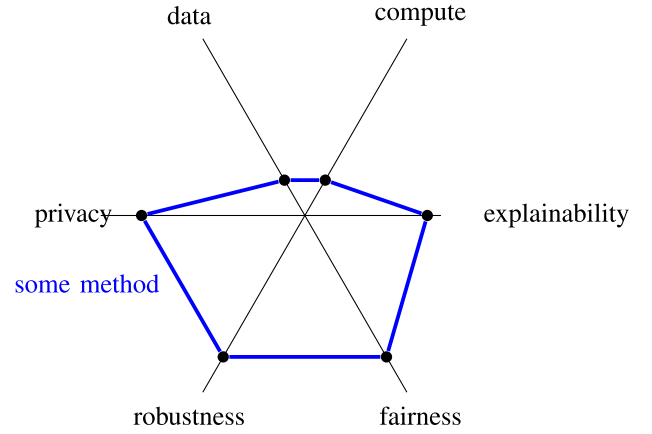


Fig. 1. Trustworthy AI adds new design criteria for ERM based methods. Besides small computational complexity and data requirements, these methods must be sufficiently explainable, privacy-friendly, fair and robust.

Fig. 1). In this article, we systematically map the design choices within ERM to the seven key requirements for trustworthy AI (KR1–KR7) outlined by the European Commission (see Table I). In the following, we will show that trustworthy AI can be engineered by rigorously designing ERM components to align with values such as human agency, technical robustness, privacy, fairness, transparency, societal well-being, and accountability. Importantly, achieving trustworthy AI goes beyond good design choices and requires interdisciplinary teams of developers, lawyers, ethics specialists, and other experts in assessing the requirements considering the specific use-case.

II. EMPIRICAL RISK MINIMIZATION AS AI ENGINE

Many of today's AI systems are based on ML techniques. The goal of ML is to predict some quantity of interest (its label) y from low-level measurements (its features) $\mathbf{x} = (x_1, \dots, x_d)^T$. The predictions are computed via a hypothesis map h which reads in the features of a data point and returns a prediction $\hat{y} = h(\mathbf{x})$ for its label.

A. Model

The hypothesis h is learnt or optimised based on the discrepancy between previous predictions and observed labels. The set of hypotheses that an ML method can choose from, is called the hypothesis space or model.

B. Loss

To select or learn a useful hypothesis from a model, we need a measure of the quality of the predictions obtained from a hypothesis. For this purpose, ML methods use loss functions $L((\mathbf{x}, y), h)$ to obtain a quantitative measure prediction errors.

C. Risk

The ultimate goal of ML is to learn a hypothesis $\hat{h} \in \mathcal{H}$ that incurs a small loss when predicting the label of any data point.

TABLE I
MAPPING ERM DESIGN COMPONENTS TO TRUSTWORTHY AI REQUIREMENTS (WITH REFERENCES TO SECTIONS)

	KR1	KR2	KR3	KR4	KR5	KR6	KR7
Data	Sec. IV-A	Sec. V-B	Sec. VI-A	Sec. VII-B	Sec. VIII-A	Sec. IX	Sec. X
Model	Sec. IV-B	Sec. V-C	Sec. VI-B	Sec. VII-C	Sec. VIII-B	Sec. IX	Sec. X
Loss	Sec. IV-C	Sec. V-A	Sec. VI-C	Sec. VII-A	Sec. VIII-C	Sec. IX	Sec. X

We can make this informal requirement more precise by interpreting data points as realisations of i.i.d. random variable (RV) with a common probability distribution $p(\mathbf{x}, y)$. This allows to define the expected loss or risk of a hypothesis

$$\bar{L}(h) := \mathbb{E}\{L((\mathbf{x}, y), h)\}. \quad (1)$$

D. Data

As the underlying probability distribution $p(\mathbf{x}, y)$ is typically unknown, we cannot directly optimise the risk (1). Instead, practical ML methods need to approximate the risk from a dataset

$$\mathcal{D} = \left\{ (\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(m)}, y^{(m)}) \right\}. \quad (2)$$

The dataset consists of data points, each characterized by features $\mathbf{x}$ and some label y . Empirical risk minimization (ERM)-based methods require a dataset to measure the usefulness of a hypothesis $h \in \mathcal{H}$ and, in turn, to train a model (learn a useful choice of model parameters).

E. Empirical Risk

Arguably, the most widely used approximation to risk (1) is the average loss or empirical risk (see Fig. 2)

$$\hat{L}(h|\mathcal{D}) := (1/m) \sum_{r=1}^m L((\mathbf{x}^{(r)}, y^{(r)}), h). \quad (3)$$

Much of statistical learning theory revolves around the study of the approximation $\hat{L}(h|\mathcal{D}) \approx \bar{L}(h)$. The approximation quality can be studied using various forms of the law of large numbers or concentration inequalities [25], [26], [27], [28].

F. ERM

ERM-based methods learn a hypothesis $\hat{h} \in \mathcal{H}$ from a hypothesis space (or model) $\mathcal{H}$ by minimizing the empirical risk $\hat{L}(h|\mathcal{D})$ as a proxy for the risk

$$\begin{aligned} \hat{h} &:= \operatorname{argmin}_{h \in \mathcal{H}} \hat{L}(h|\mathcal{D}) \\ &= \operatorname{argmin}_{h \in \mathcal{H}} \sum_{(\mathbf{x}, y) \in \mathcal{D}} L((\mathbf{x}, y), h). \end{aligned} \quad (4)$$

We obtain practical ML systems by applying optimisation methods to solve (4). Different ML methods are obtained from different design choices for the data points (their features and labels), the hypothesis space (or model), and the loss function [24, Ch. 3].

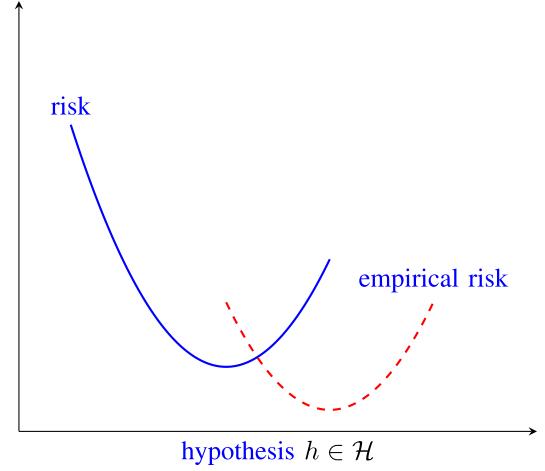


Fig. 2. ERM uses the average loss incurred on a training set to approximate the risk (or expected loss).

G. Design Choices

From a ML engineering perspective, the design choices in ERM are mainly guided by computational aspects and statistical aspects of the resulting optimisation problem (4). Computational aspects include the number of arithmetic operations required by a ML method. Statistical aspects include the generalization error $\bar{L}(\hat{h}) - \hat{L}(\hat{h}|\mathcal{D})$ of the learnt hypothesis and its robustness against the presence of outliers in the training set (2).

H. Generalization

While measuring the computational complexity by counting arithmetic operations is quite straightforward [29], measuring the generalization error is more challenging. Indeed, since we typically do not know the underlying probability distribution of the data points, we can only estimate the generalization performance using a validation set. The validation set consists of data points that were not used for the training set $\mathcal{D}$ in ERM (4) (see Fig. 3).

Ensuring trustworthy AI with {erm} requires not only statistical and computational optimisation but also careful design of the training data, the ML {model}, and the {lossfunc}. This article examines how deliberate choices in each of these components can help satisfy the key requirements for trustworthy AI.

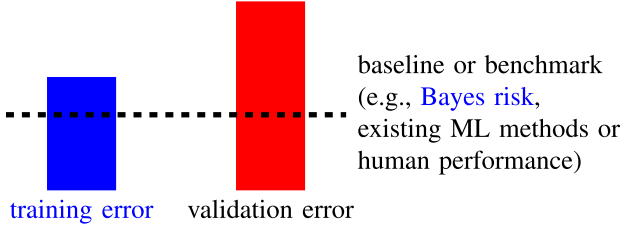


Fig. 3. We can diagnose a ML method by comparing its training error with its validation error. Ideally both are on the same level as a baseline (or benchmark error level).

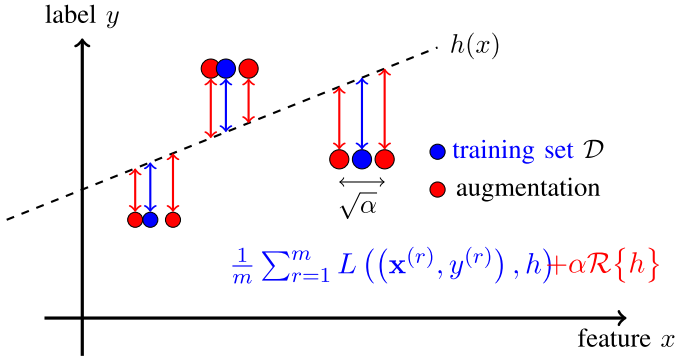


Fig. 4. Equivalence between data augmentation and loss penalization.

I. Regularization

Consider an ERM-based ML method using a hypothesis space $\mathcal{H}$ and a dataset $\mathcal{D}$ (we assume all data points are used for training). A key parameter for such ML methods is the ratio $d_{\text{eff}}(\mathcal{H})/|\mathcal{D}|$ between the (effective) model size $d_{\text{eff}}(\mathcal{H})$ and the number $|\mathcal{D}|$ of data points.¹ The tendency of the ML method to overfit increases with the ratio $d_{\text{eff}}(\mathcal{H})/|\mathcal{D}|$.

Regularization techniques reduce the ratio $d_{\text{eff}}(\mathcal{H})/|\mathcal{D}|$ through three (essentially equivalent) approaches

- 1) obtaining more data points, possibly via data augmentation,
- 2) adding a penalty term $\alpha \mathcal{R}\{h\}$ to the average loss in ERM (3),
- 3) shrinking the hypothesis space, e.g., by adding constraints on model parameters such as $\|\mathbf{w}\|_2 \leq 10$.

It can be shown that these three perspectives (corresponding to the three components data, model, and loss) on regularization are closely related [24, Ch. 7]. For example, the addition of a penalty term $\alpha \mathcal{R}\{h\}$ in ERM (3) is equivalent to ERM (3) with a pruned hypothesis space $\mathcal{H}^{(\alpha)} \subseteq \mathcal{H}$. Using a larger α typically results in a smaller $\mathcal{H}^{(\alpha)}$ [24, Ch. 7]. Furthermore, adding the penalty term $\alpha \mathcal{R}\{h\}$ is equivalent to augmenting the original training set with perturbations of its data points (see Fig. 4).

¹Arguably, the most widely used measure for the effective size of a ML model is the Vapnik–Chervonenkis (VC) dimension [25]. However, the precise definition of the model size is not relevant for our discussion.

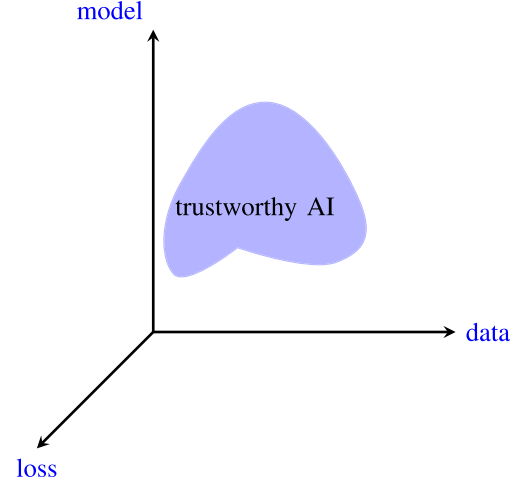


Fig. 5. ERM-based methods are defined by design choices for data, model and loss. This article discusses design choices that facilitate KRs for trustworthy AI.

III. DESIGN CHOICES AND TRUSTWORTHY AI

In the design of an AI application, there are many different design choices that can be made (see Fig. 4). While traditionally computational and statistical aspects are the most prominent guiding principles, it is crucial for the further development of AI to take into account requirements of trustworthiness to ensure that AI applications are also socially beneficial and save. The European Union has identified seven key requirements for trustworthy artificial intelligence [20]. Even though these requirements are not legally binding, their ultimate legal basis lies in the EU Charter of Fundamental Rights and, therefore, in the EU's core democratic values. To create trustworthy AI it is essential to develop an understanding of these norms. Hence, in the following, we will propose different design choices in the context of the key requirements and provide more information about their relationship to core values of the European Union (see Fig. 5). However, it is important to note that technical design choices represent only one pillar of trustworthy AI. Especially oftentimes required balancing tests of rights assessments will have to be conducted by legal experts in cooperation with the AI experts. Additionally, organizational, contractual and educational factors must be considered to ensure that AI applications are truly trustworthy.

IV. KR1 - HUMAN AGENCY AND OVERSIGHT

The basic idea of human agency and oversight [20, p.15] relates to human autonomy as the right to live one's life as one wishes (liberty) [30, Article 6], and human dignity, the fact that all persons, by virtue of being human, have an inherent, equal, and inalienable value regardless of any other factors [30, Article 1]. Human agency and oversight implement these values by requiring that ultimately a person be meaningfully involved in the process of decision-making; an AI system shall not be able to make uncontrolled decisions about people. To ensure feasibility of this oversight, AI systems need to provide for transparency, which could be seen as precondition

for agency and oversight. Transparency, however, is a separate requirement within the guidelines and will be treated separately in the following.

ERM-based methods must be designed to support user agency, ensure human oversight, and protect fundamental rights. The predictions delivered by a trained model must not result in any manipulation or undue influence that would deprive the person subjected to decisions of meaningful decision making. We must ensure safeguards to maintain human control and to prevent harmful outcomes. KR1 is closely related to fundamental rights such as dignity, freedom, and nondiscrimination [30].

A. Human Agency

Users should be able to understand, interact with, and challenge decisions based on predictions $\hat{h}(\mathbf{x})$ delivered by a trained model $\hat{h} \in \mathcal{H}$. Human agency is facilitated by using transparent models and comprehensive documentation of the training process [e.g. optimization method used to solve (3)]. The ERM design choices for data points (their features and label) and loss function must ensure that the trained model avoids any manipulation or deception of users, all of which may threaten individual autonomy [31], [32], [33].

B. Human Oversight

We must design ERM-based methods that do not compromise autonomy or cause harm. This can be implemented through various governance models that allow for varying degrees of human intervention, from direct involvement in learning cycles (monitoring gradient bases methods to solve (3)) to greater oversight of the social and ethical impacts resulting from predictions $h(\mathbf{x})$.

Next, we discuss how KR1 guides the design choice for data (training set), model and loss used in ERM (4).

C. Data

1) *Freedom of the Individual*: Guaranteeing individual freedom requires that individuals, especially those at risk of exclusion, have equal access to the benefits and opportunities that AI can offer. In this regard, KR1 requires that the training set in (3) is curated with diversity and inclusion in mind. Biases in data collection or labeling may disproportionately affect certain groups, potentially limiting their autonomy or reinforcing discriminatory patterns. Unbiased representation of subpopulations in the dataset $\mathcal{D}$ used by ERM (4) is essential to avoid manipulation, deception, or discrimination of individuals and to protect their autonomy and freedom of choice.

2) *Respect for Human Dignity*: Learning personalized model parameters for recommender systems allows supplying customised suggestions to users, referred to as microtargeting. This can be useful as it can help users to find suitable contents or products. However, microtargeting can also increase addictive user behavior or even emotional manipulation of larger user groups [17], [34], [35], [36]. KR1 rules out certain design choices for the labels of data points to defuse microtargeting. In

particular, we must avoid using the mental and psychological characteristics of a user as the label. KR1 also rules out loss functions that can be used to train predictors of psychological characteristics.

3) *Continuous Monitoring*: In their simplest form, ERM-based methods involve a single training phase, i.e. they solve (3) using some numerical optimisation method [37], [38]. The use of a single training phase is only useful if the data generation is stationary, e.g., if it can be well approximated by an i.i.d. assumption. For many ML applications, this assumption is only realistic if the training set is limited to a sufficiently short time period [39], [40]. It is crucial to continuously compute a validation error on a current validation set used to diagnose the whole ML system (see [24, Sec. 6.6]). Based on the diagnosis, the model parameters can be updated (retrained) by using a fresh training set for ERM.

D. Model

Human agency and oversight can be facilitated by relying on simple models such as linear models with few features or decision trees with small depth. Defining what makes a model “simple” is challenging. However, a more rigorous understanding of model simplicity can be developed through quantitative measures of their explainability (or interpretability). Section VII constructs measures of the subjective explainability of a trained model $\hat{h} \in \mathcal{H}$. Generally speaking, we refer to a model as simple if a nonexpert can (to some degree) understand how the features of a data point relate to the prediction $h(\mathbf{x})$.

While the internal logic of simple models (such as linear classifiers) may be transparent, malicious actors may repurpose them in harmful or unintended ways. These attacks on simple models are difficult to foresee at design time. Thus, while the use of simple (or interpretable) models supports oversight and auditing, it must be complemented by broader socio-technical assessments and usage controls [41], [42].

E. Loss

The choice of the loss function in ERM (4) should favour hypothesis maps $h \in \mathcal{H}$ that are consistent with fundamental rights. For example, including a penalty term in the loss function can force the trained model to yield predictions that are invariant across different mental states of the same user, as even though those would not be included in the initial data collection, it is still possible that those could be accidentally inferred from user behaviour. We can also explicitly incorporate domain expertise of psychologists to penalize predictions that would recommend harmful content to social media users [43].

1) *Interpretable Loss Function*: To facilitate human oversight, we should use a loss function that is intelligible to the user. For example, consider a user without formal training or education in ML. Here, using the 0/1 loss might enable human oversight more efficiently compared with using the logistic loss [24, Sec. 2.3.2].

2) *Incorporate Human-Centric Objectives*: We can choose a loss function that includes penalty terms that reflect human-centric values such as fairness (see Section VIII) or transparency (see Section VII). The idea is to penalize a hypothesis h that delivers predictions $h(\mathbf{x})$ that contradict these goals.

3) *Penalizing Unethical Outcomes*: The loss function in ERM can be tailored to penalize a prediction $h(\mathbf{x})$ that is considered to be unethical. For instance, we could assign a very large loss value to a prediction that results in presenting fake news to a user.

V. KR2 - TECHNICAL ROBUSTNESS AND SAFETY

ERM-based ML methods should work reliably even when things go wrong. They must be safe in real-world conditions where data might be messy, models might be slightly wrong, and computing systems might fail. This is the idea behind *Key Requirement 2 (KR2)* [20, p.16], which supports important rights in the EU Charter such as the right to life [30, Article 2], protection of health [30, Article 3], and protection of personal data [30, Article 8].

In this section, we look at what can go wrong in ERM-based systems and how to design them to stay safe and reliable.

Where Do Problems Come From? To solve an ERM problem like (3), we run optimization algorithms on real computers. These computers are not perfect [44], [45]

- 1) The code might have bugs.
- 2) Hardware can crash or lose power.
- 3) Communication links might fail.
- 4) There is always some rounding error in floating-point calculations.

Even if the computation works fine, the training data $\mathcal{D}$ might be noisy or tampered with

- 1) Sensors deliver noisy measurements [46].
- 2) Attackers can inject fake or misleading data [47], [48].

In practice, the data used for training is typically a random sample drawn from an unknown underlying distribution [24], [25]. As a result, each training set may differ slightly. A robust model should maintain strong performance despite such variability in the training data.

A. Stability Helps Generalization

A central concept in learning theory is that if a learning algorithm exhibits only minor changes in its output when the training set is perturbed slightly, it is likely to generalize well to unseen data [49], [50], [51]. In other words, stable methods often perform well on new, unseen data.

While many forms of data variation are small and random, real-world datasets often contain more severe forms of contamination. In particular, the presence of even a small number of extreme outliers—arising from sensor malfunctions, data entry errors, or adversarial manipulation—can severely compromise the reliability of a learning algorithm. This vulnerability poses a critical challenge in safety-sensitive domains such as medical diagnostics or autonomous systems.

Robust statistics offers a principled framework for addressing this issue. A key concept in this field is the *breakdown*

point, which serves as a formal robustness metric for statistical estimators. Specifically, the breakdown point quantifies the largest proportion of contaminated data an estimator can tolerate before it yields arbitrarily incorrect results. For example.

- 1) The sample mean has a breakdown point of zero; a single outlier can render it invalid.
- 2) The sample median, by contrast, has a breakdown point of 50%, making it substantially more robust [52].

What Can Go Wrong? Robustness means the model still performs well when things go slightly wrong. These problems can happen in three places.

- 1) *Data*: Noise, adversarial samples, or changes in the data distribution.
- 2) *Model*: Changes in architecture, unstable parameters, or software updates.
- 3) *Loss Function*: Poorly chosen objective, or adversarial manipulation of the loss.

We describe this by allowing changes to the training set, model class, and loss function

$$(\tilde{\mathcal{D}}, \tilde{\mathcal{H}}, \tilde{L}) \in \mathcal{U}.$$

This leads to a modified ERM objective

$$\tilde{h} = \operatorname{argmin}_{h \in \tilde{\mathcal{H}}} \frac{1}{|\tilde{\mathcal{D}}|} \sum_{(\mathbf{x}, y) \in \tilde{\mathcal{D}}} \tilde{L}(\mathbf{x}, y, h). \quad (5)$$

We can then ask: how different is the model we get from (3) compared with the one from (5)? This gives us a way to measure robustness.

What Can Go Wrong (KR2)

To make learning systems safe and robust, we need to consider.

- **Data problems**: sensor noise, data poisoning, distribution shift
- **Model problems**: unstable architectures, numerical errors
- **Loss problems**: poor objectives, adversarial manipulations

B. Loss

This section discusses specific choices (constructions) for the loss function in ERM (4) so that its solutions are close to the perturbed ERM (5). In particular, we consider uncertainty sets that contain a single choice of model $\mathcal{H}$ and loss function L but different perturbed datasets $\tilde{\mathcal{D}}$. Thus, throughout this section, we assume that (4) and (5) use the same $\mathcal{H}$ and L .

1) *Robust Statistics*: A well-known example of a loss function that improves the robustness of ERM is the Huber loss. Using the Huber loss instead of the squared error loss in (4) makes the resulting method significantly more robust (or insensitive) against the presence of outliers in the training set $\mathcal{D}$ [24, Ch. 3]. In fact, in the special case of learning a constant predictor (that is, estimating the mean of a sample), minimizing the empirical risk under the squared error loss yields the sample mean. In

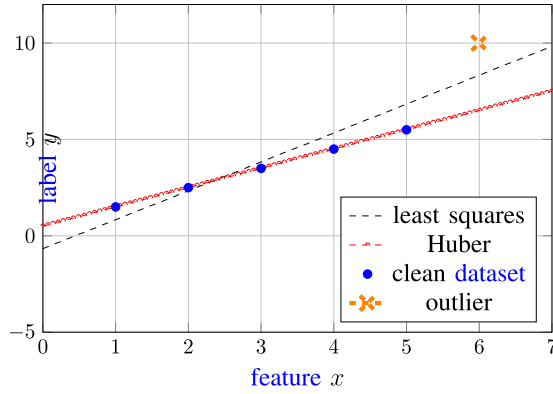


Fig. 6. Effect of using either squared error or Huber loss on learning the parameters of a linear model. The model parameters learnt by minimizing the average Huber loss seem to be more robust against the presence of an outlier.

contrast, minimizing the empirical risk under the Huber loss results in a value closer to the sample median, particularly as the Huber threshold parameter tends to zero. Fig. 6 depicts a toy dataset along with two linear models, one trained by minimizing the average squared error loss and another one by minimizing the average Huber loss.

2) *Adversarial Loss*: A principled construction of robust loss functions is based on replacing ERM (4) with an adversarial (or worst-case) variant [53], [54]

$$\hat{h} = \operatorname{argmin}_{h \in \mathcal{H}} \sup_{\tilde{\mathcal{D}} \in \mathcal{U}} \sum_{(\mathbf{x}, y) \in \tilde{\mathcal{D}}} L((\mathbf{x}, y), h). \quad (6)$$

A rapidly growing body of work examines various instances of (6) obtained for different constructions of the uncertainty set $\mathcal{U}$ [55], [56], [57], [58]. In the following, we show how a special case of (6) is equivalent to (4) for a suitable choice L' for the loss function.

A common construction of the uncertainty set in (6) is to perturb separately the features and labels of the data points in $\mathcal{D}$ [55], [57]. Thus, the uncertainty set is decomposed into a separate sets $\mathcal{U}^{(\mathbf{x}, y)}$, one for each data point $(\mathbf{x}, y)$ in $\mathcal{D}$. The adversarial ERM (6) then becomes [57]

$$\hat{h} = \operatorname{argmin}_{h \in \mathcal{H}} \sum_{r=1}^m \underbrace{\sup_{(\tilde{\mathbf{x}}, \tilde{y}) \in \mathcal{U}^{(\mathbf{x}^{(r)}, y^{(r)})}} L((\tilde{\mathbf{x}}, \tilde{y}), h)}_{\text{robust loss } L'((\mathbf{x}^{(r)}, y^{(r)}), h)}. \quad (7)$$

Note that the robust loss function L' in (7) depends both on the original choice for the loss function in (4), as well as the uncertainty set $\mathcal{U}$ in (7).

Next, consider a modification of (7) where we perturb only the features of the data points but leave their labels untouched [59]. This modification uses an uncertainty set $\mathcal{U}^{(\eta)}$ which is parametrized by a perturbation strength η and consists of datasets $\tilde{\mathcal{D}} = (\tilde{\mathbf{X}}, \mathbf{y})$ with feature matrix

$$\begin{aligned} \tilde{\mathbf{X}} &= \mathbf{X} + (\mathbf{u}^{(1)}, \dots, \mathbf{u}^{(d)}) \\ &:= (\tilde{\mathbf{x}}^{(1)}, \dots, \tilde{\mathbf{x}}^{(m)})^T := (\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)})^T \end{aligned}$$

$$\text{with } \|\mathbf{u}^{(j)}\|_1 \leq \eta. \quad (8)$$

Note that, unlike in (7), the construction (8) couples the features of different data points in $\mathcal{D}$. Thus, instead of maximizing separately over possible perturbations for each data point as in (7), we need to study the worst-case perturbation of the entire dataset

$$\hat{h} = \operatorname{argmin}_{h \in \mathcal{H}} \sup_{\tilde{\mathcal{D}} \in \mathcal{U}} \sum_{r=1}^m L((\tilde{\mathbf{x}}^{(r)}, y), h). \quad (9)$$

Consider ERM obtained for a linear model and the absolute error loss. Here, it can be shown that (9) is equivalent to ERM with the robust loss $L' = |y - h(\mathbf{x})| + \eta \|\mathbf{w}\|_1$ [59, Thm. 14.9.].

Recent work also studies uncertainty sets $\mathcal{U}$ consisting of perturbed datasets $\tilde{\mathcal{D}}$ with an empirical distribution $\tilde{\mathbb{P}}$ close to the empirical distribution $\mathbb{P}$ of $\mathcal{D}$

$$\mathcal{U}^{(\eta)} := \{\tilde{\mathcal{D}} : W(\tilde{\mathbb{P}}, \mathbb{P}) \leq \eta\}. \quad (10)$$

Here, $W(\tilde{\mathbb{P}}, \mathbb{P})$ denotes the Wasserstein distance between $\tilde{\mathbb{P}}$ and $\mathbb{P}$ [58].

Consider an adversarial ERM (6) with uncertainty set (10) and a loss function L that is Lipschitz continuous with modulus α . It can then be shown that (6) is equivalent to ERM with a certain robust loss function [58, Theorem 4].

For a binary classification, the authors of [60] study a robust loss of the form

$$L((\mathbf{x}, y), h) = \begin{cases} 1 & \text{if } h(\mathbf{x}') \neq y \text{ for some } \mathbf{x}' \in \mathcal{U}_{\mathbf{x}} \\ 0 & \text{otherwise.} \end{cases} \quad (11)$$

Here, $\mathcal{U}_{\mathbf{x}}$ is some robustness region. Note that (11) reduces to the basic 0/1 loss for the choice $\mathcal{U}_{\mathbf{x}} = \{\mathbf{x}\}$ [24].

C. Data

Instead of choosing a robust loss function L in ERM (4), we can construct the training set in (4) to make its solutions more robust. A widely studied approach is adversarial training, i.e., including adversarially perturbed data points in the training set $\mathcal{D}$ [54], [61], [62].

An opposite approach to adversarial training is to prune a given dataset using some form of outlier detection [63]. The training set is then obtained by the remaining data points that have not been declared as outliers.

However, it can be challenging to distinguish outliers from natural perturbations due to the sampling from a true underlying probability distribution [64], [65].

The fundamental limits for outlier removal techniques can be studied using a malicious noise model [66]

$$\mathbf{z}^{(r)} = \begin{cases} \tilde{\mathbf{z}}^{(r)} & \text{if } b^{(r)} = 1 \\ \mathbf{e}^{(r)} & \text{otherwise,} \end{cases} \quad (12)$$

$$\text{with } b^{(r)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{B}(p_e), \tilde{\mathbf{z}}^{(r)} \stackrel{\text{i.i.d.}}{\sim} p(\mathbf{x}, \mathbf{y}). \quad (13)$$

Here, the outlier $\mathbf{e}^{(r)}$ can be chosen arbitrarily (maliciously), even taking into account the current state of the optimisation method used to solve (4). Consider an ERM method for binary classification, that returns a hypothesis $\hat{h}$ with expected 0/1 loss

$\mathbb{E}\{L(\mathbf{z}, \hat{h})\}$. To allow for the existence of an ERM method achieving $\mathbb{E}\{L(\mathbf{z}, \hat{h})\} < \varepsilon$, the maximum fraction of outliers that can be tolerated is upper bounded by $\varepsilon/(1 + \varepsilon)$ [66].

D. Model

We can define and measure the robustness of ML using different notions of continuity of the learnt hypothesis $\hat{h}$. Besides the basic qualitative notion of continuity, we can also use Lipschitz continuity to obtain a quantitative measure of robustness [67]. Note that Lipschitz continuity requires both the domain and the range of the hypothesis map $\hat{h}$, to be a metric space.

An obvious way to ensure the robustness of ERM is to use a model $\mathcal{H}$ that contains only Lipschitz continuous hypothesis maps $\mathcal{H}$. Recent work has shown that ERM provides a Lipschitz continuous hypothesis if the model $\mathcal{H}$ is sufficiently large [68].

Instead of Lipschitz continuity, the authors of [69] use the concept of local and global robustness for multiclass classification problems. Here, data points have a label $y \in \mathcal{Y} := \{1, \dots, K\}$ and the goal is to learn a classifier $h(\mathbf{x}) = (h_1(\mathbf{x}), \dots, h_K(\mathbf{x}))^T$ which is used to classify a data point as $\hat{y} = \arg\max_{c \in \{1, \dots, K\}} h_c(\mathbf{x})$.

A classifier $h(\mathbf{x})$ is then defined as ε -locally robust at feature vector $\mathbf{x}$ if it classifies $\hat{y} = \hat{y}'$ for every data point with features $\mathbf{x}'$ such that $\|\mathbf{x} - \mathbf{x}'\|_2 \leq \varepsilon$ [69]. Note that if we require this to hold for every $\mathbf{x}$, the classifier must be trivial (providing the same label value for each data point). To obtain a useful notion of global robustness, the authors of [69] introduce an auxiliary label value that signals when the classifier is not locally robust.

VI. KR3 - PRIVACY AND DATA GOVERNANCE

ERM-based methods must ensure protection against unauthorised access to and misuse of personal data. Such protection is critical in the face of concrete threats such as membership inference attacks, which attempt to infer whether a particular individual's data were used in training [70]. Data protection and privacy protection are typically implemented as part of a broader data governance framework [71], designed to ensure the quality, integrity, and lawful processing of personal data.

A. Data Governance

Many applications of ERM use data points that represent individuals, often capturing behavioral patterns, preferences, or sensitive medical information. Especially for such personal data, data governance must cover the entire AI lifecycle, from collection and training to model deployment and secure deletion. Effective governance includes data lineage, model validation (e.g., verifying factual accuracy), and documenting data sources [72].

Legal frameworks such as the General Data Protection Regulation mandate safeguards such as data protection impact assessments (DPIAs) and, in some cases, the appointment of a Data Protection Officer. These mechanisms help anticipate and mitigate privacy risks throughout the system's operation [73].

1) *Measuring Privacy Leakage*: A fundamental question is how much private information can leak from the output of an ERM-based system. Formally, we can model such a system as a function $\mathcal{A}$ that maps a training set $\mathcal{D}$ to an output $\mathcal{A}(\mathcal{D})$, such as the trained parameters $\hat{\mathbf{w}}$ or a prediction $\hat{h}(\mathbf{x})$. *Membership inference attacks* exploit this map to determine whether a particular data point was used during training—potentially violating privacy even if the model output seems innocuous.

2) *Privacy Protection and Noninvertibility*: To prevent such leakage, it must be computationally infeasible to infer sensitive components of any data point from $\mathcal{A}(\mathcal{D})$. Let each data point be represented by features $\mathbf{x}$, a label y , and a sensitive attribute s (e.g., a medical diagnosis).

Privacy Protection that the map $\mathcal{A}$ be sufficiently noninvertible—meaning not just that $\mathcal{A}(\mathcal{D})$ lacks an inverse (in a strict mathematical sense), but that even sophisticated inference attacks cannot reliably reconstruct s or deduce the presence of a specific data point. This notion is tightly linked to *Plausible Deniability*: a property whereby the inclusion of a specific data point in $\mathcal{D}$ cannot be confirmed or denied with high confidence based on the observed output.

3) *Differential Privacy (DP)*: A widely used approach to make an ERM-based method sufficiently noninvertible is to introduce some randomness or noise. Examples of such randomness include adding noise to the output and selecting a random subset of $\mathcal{D}$. The map $\mathcal{A}$ then becomes stochastic and, in turn, the output $\mathcal{A}(\mathcal{D})$ is then characterized by a probability distribution $\text{Prob}\{\mathcal{A}(\mathcal{D}) \in \mathcal{S}\}$ for all sets $\mathcal{S}$ within a well-defined collection of measurable sets [74].

DP measures the noninvertibility of a stochastic algorithm $\mathcal{A}$ by the similarity of the probability distributions obtained for two datasets $\mathcal{D}, \mathcal{D}'$ that are considered neighbouring or adjacent [75], [76]. Typically, we consider $\mathcal{D}'$ to be adjacent to $\mathcal{D}$ if it is obtained by modifying the features or label of a single data point in $\mathcal{D}$. In general, the notion of neighbouring datasets is a design choice used in the formal definition of differential privacy.

Definition 1 (Adapted from [76]): An ERM-based method $\mathcal{A}$ is (ε, δ) -differential privacy if for any measurable set $\mathcal{S}$ and any two neighboring datasets $\mathcal{D}, \mathcal{D}'$

$$\text{Prob}\{\mathcal{A}(\mathcal{D}) \in \mathcal{S}\} \leq \exp(\varepsilon) \text{Prob}\{\mathcal{A}(\mathcal{D}') \in \mathcal{S}\} + \delta. \quad (14)$$

By limiting the influence of any single data point on the output, differential privacy ensures a robust defense against membership inference and supports plausible deniability. This makes it a key enabler of KR3-compliant AI systems.

B. Data

A simple way to implement privacy protection in ERM-based methods is through careful selection of the features used to characterize data points [77], [78]. The goal is to select features that are relevant to the learning task while ensuring they do not reveal excessive information about sensitive attributes. Note that it is typically not sufficient to just remove sensitive attributes from raw data points to ensure privacy protection. Indeed, these sensitive attributes might be recoverable from the other features of a data point.

A straightforward approach to implementing privacy protection within ERM-based methods involves the deliberate selection of features used to characterize each data point [77], [78]. The aim is to retain only those features that are relevant for the learning task, while minimizing the risk of disclosing sensitive information. However, it is typically insufficient to ensure privacy protection by merely excluding explicit sensitive attributes from the input data. This is because such attributes can often be inferred indirectly through other correlated features—known as proxy variables—that act as functional or statistical surrogates. Therefore, effective privacy protection requires a more holistic view that accounts for the interactions between features, and may involve additional techniques such as adversarial training, regularization, or differential privacy to mitigate leakage from correlated attributes.

There is an inherent trade-off between privacy protection and the resulting statistical accuracy. In fact, we can trivially achieve perfect privacy protection by not using any property of a data point as a features. However, this extreme case of maximum privacy protection comes at the cost of a lower quality of the predictions delivered by (the hypothesis learnt from) ERM.

1) *Private Feature Learning*: In many practical scenarios, it is challenging to manually select features that achieve an appropriate trade-off between privacy protection and predictive performance.² A data-driven approach to this problem is to learn a feature map $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$ that transforms raw features into representations $\mathbf{z} = \Phi(\mathbf{x})$ which preserve task-relevant information while obscuring sensitive attributes.

The goal is to learn Φ such that it enables accurate prediction of the label y , while simultaneously hindering inference of the private attribute s . We now outline two approaches to this form of private feature learning, which differ primarily in how they quantify the predictability of s and y . Both approaches rely on a probabilistic model for the data points in $\mathcal{D}$, treating them as realizations of i.i.d. random variables (RVs).

2) *Privacy Funnel*: One principled approach is based on mutual information, a well-established information-theoretic quantity that captures the statistical dependence between random variables. The MI $I(s; \Phi(\mathbf{x}))$ measures how much information the transformed features $\Phi(\mathbf{x})$ reveal about the sensitive attribute s . A small value of $I(s; \Phi(\mathbf{x}))$ indicates strong privacy protection, as it implies that s is difficult to infer from $\Phi(\mathbf{x})$.³

Similarly, the mutual information $I(y; \Phi(\mathbf{x}))$ quantifies the extent to which the representation retains information relevant for predicting the label y . Ideally, one seeks a representation that minimizes $I(s; \Phi(\mathbf{x}))$ while maintaining a sufficiently large $I(y; \Phi(\mathbf{x}))$.

This trade-off is formalized by the privacy funnel [80, (2)]

$$\min_{\Phi(\cdot)} I(s; \Phi(\mathbf{x})) \quad \text{subject to} \quad I(y; \Phi(\mathbf{x})) \geq R \quad (15)$$

²For instance, consider data points represented by high-dimensional raw image data, where each pixel's color intensity constitutes a feature.

³Connections between mutual information-based privacy metrics and differential privacy have been explored, e.g., in [79].

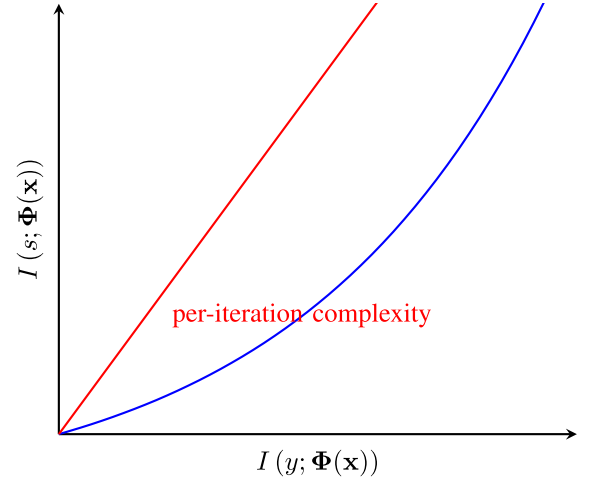


Fig. 7. Solutions of the privacy funnel (15) trace out (for varying constraint R in (15)) a curve in the plane spanned by the values of $I(s; \Phi(\mathbf{x}))$ (measuring the privacy leakage) and $I(y; \Phi(\mathbf{x}))$ (measuring the usefulness of the transformed features for predicting the label).

where R is a lower bound on the retained task-relevant information. Fig. 7 qualitatively illustrates solutions to (15) for different values of R .

Note that the exact evaluation of mutual information is typically impossible as this requires knowing the probability distribution underlying the data generation. In practice, we need to estimate mutual information from data using statistical inference techniques. These techniques rest on different modeling assumptions and, in turn, introduce approximation errors that must be considered when designing practical systems based on the privacy funnel.

3) *Private Linear Feature Learning via Reconstruction Error*: The privacy funnel (15) uses the mutual information $I(s; \Phi(\mathbf{x}))$ to quantify the privacy leakage of a feature map $\Phi(\mathbf{x})$. An alternative measure of privacy leakage is the minimum reconstruction error $s - \hat{s}$. The reconstruction $\hat{s}$ is obtained by applying a map $r(\cdot)$ to the transformed features $\Phi(\mathbf{x})$. If the joint probability distribution $p(s, \mathbf{x})$ is a multivariate normal distribution and the $\Phi(\cdot)$ is a linear map (of the form $\Phi(\mathbf{x}) := \mathbf{F}\mathbf{x}$ with some matrix $\mathbf{F}$), then the optimal reconstruction map $r(\cdot)$ is again linear [81].

We would like to find the linear feature map $\Phi(\mathbf{x}) := \mathbf{F}\mathbf{x}$ such that for any linear reconstruction map $\mathbf{r}$ (resulting in $\hat{s} := \mathbf{r}^T \mathbf{F}\mathbf{x}$) the expected squared error $\mathbb{E}\{(s - \hat{s})^2\}$ is large. The minimum expected squared error loss

$$\varepsilon(\mathbf{F}) := \min_{\mathbf{r} \in \mathbb{R}^{d'}} \mathbb{E}\{(s - \mathbf{r}^T \mathbf{F}\mathbf{x})^2\} \quad (16)$$

measures the level of privacy protection offered by the new features $\mathbf{z} = \mathbf{F}\mathbf{x}$. The larger the value $\varepsilon(\mathbf{F})$, the more privacy protection is offered. It can be shown that $\varepsilon(\mathbf{F})$ is maximized by any matrix $\mathbf{F}$ whose rows are orthogonal to the cross-covariance vector $\mathbf{c}_{\mathbf{x},s} := \mathbb{E}\{\mathbf{x}s\}$, i.e., whenever $\mathbf{F}\mathbf{c}_{\mathbf{x},s} = \mathbf{0}$. A particular choice of $\mathbf{F}$ that satisfies this orthogonality condition is

$$\mathbf{F} = \mathbf{I} - (1 / \|\mathbf{c}_{\mathbf{x},s}\|_2^2) \mathbf{c}_{\mathbf{x},s} \mathbf{c}_{\mathbf{x},s}^T. \quad (17)$$

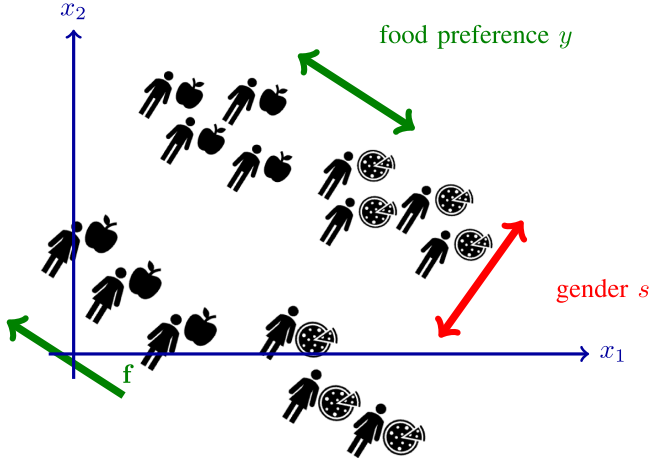


Fig. 8. Toy dataset $\mathcal{D}$ whose data points represent customers, each characterized by features $\mathbf{x} = (x_1, x_2)$. These raw features carry information about a sensitive attribute s (gender) and the label y (food preference) of a person. The scatterplot suggests that we can find a linear feature transformation $\mathbf{F} := \mathbf{f}^T \in \mathbb{R}^{1 \times 2}$ resulting in a new feature $z := \mathbf{F}\mathbf{x}$ that does not allow to predicts, while still allowing to predict y .

Fig. 8 illustrates a dataset for which we want to find a linear feature map $\mathbf{F}$ such that the new features $\mathbf{z} = \mathbf{F}\mathbf{x}$ do not accurately predict a sensitive attribute.

4) *Sufficient Statistics:* So far, we have discussed privacy protection in the sense of not allowing the prediction of sensitive attributes. In some applications, it may not be clear what a sensitive attribute is. Nevertheless, we want to minimize any potential privacy leakage. To implement such a data minimization principle, we can use the concept of a sufficient statistic [82], [83]. For this purpose, we assume that data points are obtained as i.i.d. samples from a probability distribution $p(\mathbf{x}; \mathbf{w})$ which is parametrised by model parameters $\mathbf{w}$.

ERM-based methods can be interpreted as methods for estimating the true underlying $\mathbf{w}$ of the probability distribution $p(\mathbf{x}; \mathbf{w})$. A statistic $\mathbf{z} = \Phi(\mathbf{x})$, with some map $\Phi(\cdot)$, is sufficient for the parameter $\mathbf{w}$ if the conditional probability distribution of $\mathbf{x}$, given the statistic $\mathbf{z} = \Phi(\mathbf{x})$, does not depend on the model parameters $\mathbf{w}$.

Whenever we have identified a sufficient statistic for the probabilistic model $p(\mathbf{x}; \mathbf{w})$, we can safely discard the original raw features and instead use the sufficient statistic $\mathbf{z} = \Phi(\mathbf{x})$ as the new features. Of particular interest are sufficient statistics which are minimal in the sense that any other sufficient statistic is determined from the known value of a minimal sufficient statistic [81].

C. Model

The feature learning techniques from Section VI-A above can also be implemented as a design choice for the model used in ERM. In fact, we can think of the linear feature learning map as a preprocessing step within a hypothesis map.

Recall that privacy protection of an ERM-based method $\mathcal{A}(\mathcal{D})$ is determined by its noninvertibility. Let us now illustrate the effect of the choice of the model $\mathcal{H}$ on the noninvertibility

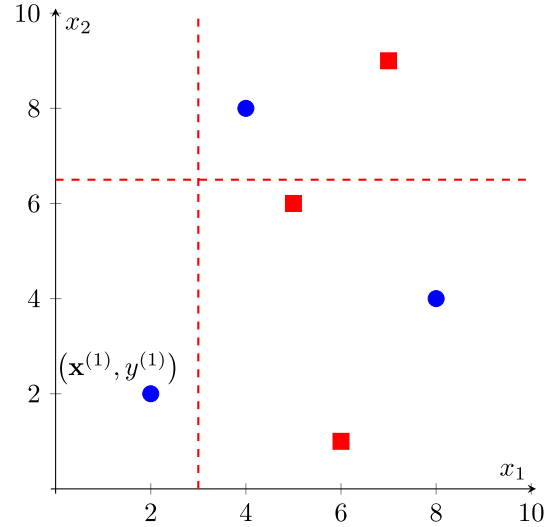


Fig. 9. Scatterplot of a dataset $\mathcal{D}$ along with the decision boundary of a decision tree $\hat{h}$ trained via ERM on $\mathcal{D}$. one of the decision regions contains a single data point from the training set which could allow an adversary to infer the label $y^{(1)}$ from the predictions $\hat{h}(\mathbf{x})$ obtained for features near-by $\mathbf{x}^{(1)} \times \{(1)\}^S$.

of $\mathcal{A}(\mathcal{D})$. Fig. 9 depicts a toy along with the decision regions of the hypothesis $\hat{h}$ learnt with ERM (4) using a decision tree model $\mathcal{H}$ [24, Ch. 3].

Note that one of the decision regions depicted in Fig. 9 contains a single data point, denoted $(\mathbf{x}^{(1)}, y^{(1)})$, from $\mathcal{D}$. Thus, if we have a sufficiently accurate estimate for the features $\mathbf{x}^{(1)}$, we can infer the label $y^{(1)}$ by observing the predictions delivered by $\hat{h}$ for features near-by $\mathbf{x}^{(1)}$. To avoid such a model inversion attack, we should use a more shallow decision tree model such that each resulting decision region contains at least a minimum number of data points from $\mathcal{D}$.

D. Loss

We can also ensure privacy protection in ERM-based AI systems by appropriate design choices for the loss function L (4). As a (not very useful) extreme case, consider a constant loss function $L((\mathbf{x}, y), \mathbf{w}) = 0$. Here, the hypothesis learnt by ERM (4) is completely unrelated to the data points in the training set $\mathcal{D}$ and, in turn, does not carry any information about them (in particular, their sensitive attributes). This maximal privacy protection comes at the cost of learning a generally useless hypothesis.

A less trivial construction of a privacy-friendly loss function is studied in [84]. Given the loss function of a potentially nonprivate ERM-based method, we simply add a random linear function to the objective function in (4). The authors of [84] then study differential privacy guarantees (14) guaranteed by using the randomly perturbed ERM.

VII. KR4 - TRANSPARENCY

Transparency [20, p.18] is to enable a person to utilise their right to take action where they believe they have been treated

unfairly. This is closely related to the right to data protection. To take actions against a potentially unlawful processing or an unjustified outcome of an ERM-based AI system, a user must have sufficient information to understand how the processing took place or how a decision was reached.

A. Traceability

The design choices (and underlying business models) for ERM-based AI systems must be documented. This includes the source of the (data points in the) training set, the model, the loss function used in (3) [72], [85]. In addition, the documentation should also include details of the optimisation method used (3). This documentation could include records of the current model parameters with a time-stamp (“logging”).

B. Explainability

Explainability is a central pillar of trustworthy AI, grounded in the principle that individuals should be able to understand the decisions made about them, particularly so they can respond or seek redress in cases of unfair treatment. Various explainability techniques have been proposed in the literature, including LIME [86] and SHAP [87]. LIME approximates a model locally around one instance with a simple model and shows which features contributed most to that instance’s prediction. SHAP on the other hand computes a contribution score (SHAP value) for each feature of each instance, based on how much it changes the model output when included vs excluded in all possible combinations. One possible downside of both approaches could still be the intelligibility of such found explanations for the data subject.

The recent court case C-203/22 [88] also highlighted the potential of counterfactual explanations to clarify automated decision-making. Counterfactual explanations were discussed by legal scholars early on [89] and can provide a clear picture of what would need to change in the circumstances of the person in order for a decision made about the person to change. However, there is no universally accepted metric for explainability so far and it is more than likely that several approaches can be used depending on the individual scenario.

Our proposed explainability measure differs from these approaches in so far as it provides a quantifiable metric for explainability. This makes it especially valuable in domains such as healthcare, where expert users can interpret and compare their own predictions with those of the model.

We adopt a definition based on the notion of simulatability [90], [91], [92], [93], [94].

A trained ML model c is explainable to a user u if its predictions align with the users own predictions.

We formalize this idea by measuring the (dis-)agreement between model predictions and user expectations on an unlabeled test set $\mathcal{D}^{(\text{test})} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m_t)}\}$

$$E^{(u)}(h) := \frac{1}{m_t} \sum_{\mathbf{x} \in \mathcal{D}^{(\text{test})}} L'(\mathbf{x}, u(\mathbf{x}), h(\mathbf{x})) \quad (18)$$

where $u(\mathbf{x})$ denotes the pseudolabel provided by user u for input $\mathbf{x}$.⁴

This explainability measure has several attractive properties. First, it is actionable: the term $E^{(u)}(h)$ can be incorporated directly as a regularizer during training [95]. Second, it is user-specific: different users may yield different values for $E^{(u)}(h)$ [96]. Third, by choosing an appropriate test set $\mathcal{D}^{(\text{test})}$, it supports both local and global forms of explainability.

C. Post-Hoc Versus Intrinsic Explainability

Explainability-enhancing techniques can be broadly categorized into two types. Post-hoc methods aim to generate explanations after model training, such as feature importance scores or saliency maps. In contrast, intrinsically explainable models—such as sparse linear models or shallow decision trees—are designed to be transparent by construction [97]. A principled alternative is to incorporate the explainability loss (18) directly into training to regularize the learned model toward higher user agreement [95].

D. Objectivity Versus Subjectivity

A major challenge in satisfying KR4 is the inherently subjective nature of explainability [95], [96]. For example, a linear model trained to predict disease risk from medical features might be explainable to a clinician but unintelligible to a layperson. This variability is captured by the user-specific labeling function $u(\mathbf{x})$ in (18), which differs across users for the same input $\mathbf{x}$.

To accommodate group-level explainability—e.g., agreement with a panel of experts—we can generalize (18) as follows:

$$E^{(\mathbf{u})}(h) := \frac{1}{m_t} \sum_{\mathbf{x} \in \mathcal{D}^{(\text{test})}} L'(\mathbf{x}, \mathbf{u}(\mathbf{x}), h(\mathbf{x})) \quad (19)$$

where $\mathbf{u}(\mathbf{x}) = (u^{(1)}(\mathbf{x}), \dots, u^{(n)}(\mathbf{x}))$ collects pseudolabels from n different users (e.g., being expert panel members). For real-valued predictions, a natural choice of L' is the squared deviation between $h(\mathbf{x})$ and the average user label, i.e., $(1/n) \sum_{j=1}^n u^{(j)}(\mathbf{x})$.

E. Local Versus Global Explainability

We can measure local or global explainability by how we choose the test set $\mathcal{D}^{(\text{test})}$ in (18). For local explainability, we use feature vectors that are close to a specific input $\mathbf{x}$. This helps check how well the model agrees with the user in a small neighborhood around $\mathbf{x}$ [98]. For global explainability, we choose a test set that covers the entire input space or a representative sample of it to assess model behavior more broadly.

1) *Loss*: Consider an ERM-based method that learns a hypothesis $\hat{h}$ via solving (4). We can ensure the explainability of $\hat{h}$ by augmenting the original loss L in (4) by the measure (18) (or (18) for group-level explainability). Thus, we use (3) with a modified loss function

$$L((\mathbf{x}, y), h) + \alpha E^{(u)}(h). \quad (20)$$

⁴The authors of [95] refer to this pseudolabel as a user signal.

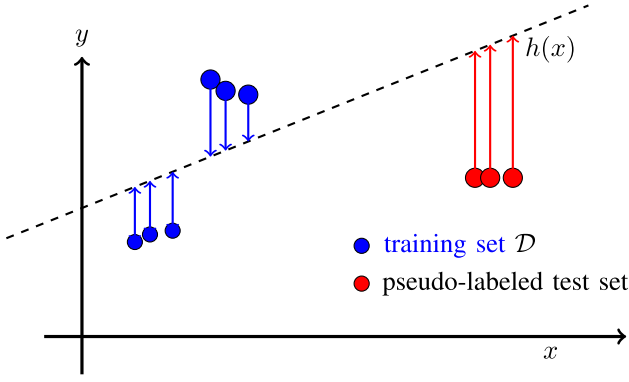


Fig. 10. We can improve simulatability (or subjective explainability) of ERM by augmenting the training set with pseudolabeled data points. These are obtained from having the user predict labels of data points in a test set.

2) *Data*: Consider that we use identical loss functions L and L' in ERM (4) and the explainability measure (18). We can then achieve the same effect as using the modified loss function (20) via augmenting the original training set (2). The augmented training set becomes

$$\mathcal{D} \cup \{(\mathbf{x}, u(\mathbf{x})) : \mathbf{x} \in \mathcal{D}^{(\text{test})}\}. \quad (21)$$

The test set $\mathcal{D}^{(\text{test})}$ may be obtained by collecting new raw data or by systematic modifications of data points in the original training set (see Fig. 10). For example, the modification may consist of constructing counterfactual examples by removing or changing important features [91].

3) *Model*: “Simple” Models: A trivial way to ensure explainability of ERM (3) is to choose a model $\mathcal{H}$ that contains only hypothesis maps incurring a small value of (18) (i.e., high explainability). However, this choice must consider the specific user’s knowledge and the construction of the test set used to compare user predictions with model predictions. For instance, a linear model may be considered explainable only if the feature space has a small dimension and the users have a basic understanding of linear functions.

Constructing Explanations: Methods for explainable AI differ not only in how they measure explainability but also in the form of explanations [96]. A widely used form of explanation is to list the most important features of a data point [98]. Another form of explanation is to use heat-maps that indicate the relative importance of image pixels [99]. Case-based reasoning uses specific data points from the training set as an explanation [100]. In general, an explanation is some function $e(\mathbf{x})$ of the features of a data point. This explanation is delivered along with the prediction to the user. Formally, this corresponds to using a hypothesis map h with structured output $h(\mathbf{x}) = (e(\mathbf{x}), \hat{y})^T$ (see Fig. 11).

VIII. KR5 - DIVERSITY, NONDISCRIMINATION, AND FAIRNESS

KR5 [20, p.18] is aligned with [30, Article 21], which prohibits discrimination based on factors such as race, gender, and religion. AI systems must treat all individuals fairly and

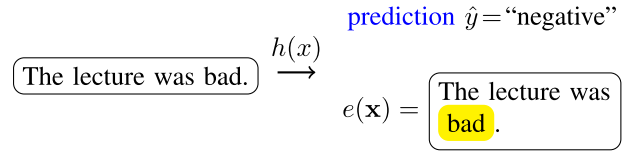


Fig. 11. We can ensure explainability of ERM-based methods by augmenting the prediction delivered by the trained model with some explanation.

inclusively, upholding their right to equality. Ensuring KR5 includes quality control of the dataset $\mathcal{D}$ used in ERM as well as the usability of interfaces for different user groups. KR5 is ultimately rooted in the inalienable value of all human beings (dignity). [20, p.18].

Consider an AI application that uses data points representing human beings. Each data point is characterized by features $\mathbf{x}$ and a sensitive attribute s . The sensitive attribute typically depends on the raw features of a data point, $s = s(\mathbf{x})$ with some map $s(\cdot)$. Examples of a sensitive attribute s are ethnicity, age, gender, or religion.⁵

A. Individual Fairness (Disparate Treatment)

Roughly speaking, a fair ERM-based method should learn a $\hat{h} \in \mathcal{H}$ that does not give undue weight to the sensitive attribute. To make this notion of fairness precise, we need a measure $d(\mathbf{x}, \mathbf{x}')$ for the similarity between data points, where features $\mathbf{x}, \mathbf{x}'$, that completely ignores their sensitive attributes [101], [102]. A fair classifier should make the same predictions for sufficiently similar data points

$$\hat{h}(\mathbf{x}) = \hat{h}(\mathbf{x}') \quad \text{whenever } d(\mathbf{x}, \mathbf{x}') \text{ is sufficiently small.} \quad (22)$$

Here, $d(\mathbf{x}, \mathbf{x}')$ is a quantitative measure of the similarity between two data points with features $\mathbf{x}, \mathbf{x}'$, respectively. The fairness requirement (22) seems to be an intuitive way of preventing *disparate treatment* [103].

B. Example: Job Platform

Consider a job platform that uses ERM to learn a hypothesis $\hat{h}$ for predicting whether a given user is suitable for a particular job opening. Each user is characterized by features $\mathbf{x} = (x_1, \dots, x_d)$ where the first entry x_1 is the age of the user. So the sensitive attribute is $s = x_1$. Fairness might require that the prediction $\hat{h}(\mathbf{x})$ does not depend on the age of the user at all [104]. We could ensure this by using a classifier that satisfies (22) with a metric $d(\mathbf{x}, \mathbf{x}')$ that does not depend on x_1 . However, the requirement (22) is insufficient if the sensitive attribute $s = x_1$ can be inferred (predicted) from the values of the remaining features $x_2, \dots, x_d$ [105], [106].

In the ML literature, a variety of quantitative measures for the fairness of a trained model have been studied and proposed $\hat{h} \in \mathcal{H}$. In the following, we briefly survey some of these measures in

⁵The definition of the sensitive attribute s is a design choice that varies by application. For instance, religion might be a sensitive attribute on a job application platform, but it could be a relevant feature in a diet planning app.

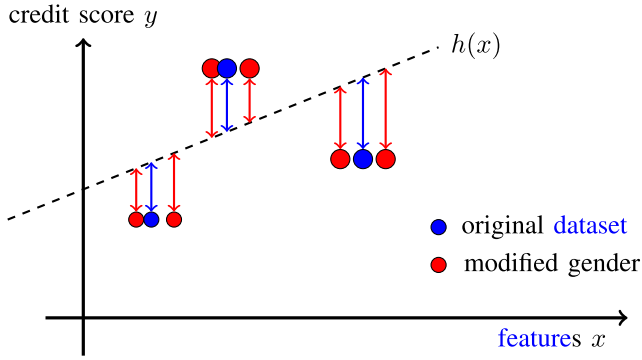


Fig. 12. We can improve fairness of a ML method by augmenting the training set using perturbations of an irrelevant feature. For example, in a credit scoring application, we might change the gender of a person while keeping the remaining features fixed.

the context of binary classification where the learnt hypothesis is used to deliver a predicted label $\hat{y} \in \{0, 1\}$.

1) *Group Fairness (Disparate Impact)*: Apart from the individual fairness constraint (22), another interpretation of fairness is to require that a trained model have similar performance across subpopulations [101], [105], [107], [108], [109], [110]. For example, we might require identical conditional risk for subsets of data points with sensitive attribute value $s^{(1)}$ and $s^{(2)}$, respectively

$$\mathbb{E}\{L((\mathbf{x}, y), \hat{h}) | s = s^{(1)}\} = \mathbb{E}\{L((\mathbf{x}, y), \hat{h}) | s = s^{(2)}\}. \quad (23)$$

Imposing (23) requires the learnt hypothesis to have the same performance (expected loss) over subpopulations of data points that have a common sensitive attribute s (e.g., “males” and “females”). The fairness requirement (23) is closely related to the notion of *disparate impact* [103].

A. Data

The training set $\mathcal{D}$ used in ERM should be carefully chosen so as not to reinforce existing discrimination. In a healthcare application, there may be significantly more training data for patients of a particular gender, resulting in models that perform best for that gender at the expense of poorer performance for the minority [75, Sec. 3.3.].

Fairness is also important for ML methods used to determine creditworthiness and, therefore, whether or not a loan should be granted [111]. Here, we need to ensure that ML methods do not discriminate against customers on the basis of ethnicity or race. To do this, we could augment data points by modifying any features that primarily reflect a customer’s ethnicity or race (see Fig. 12).

Data augmentation to enforce the fairness of the ERM is also studied in [112]. The data augmentation strategy in that article involves replacing individuals in video frames with new individuals while preserving the original motion [113], [114]. The two-step process involves: 1) tracking and segmenting the target person in the video; and 2) replacing the person with another individual by transforming key-points and poses.

Fair Data Collection: The fairness of ERM-based methods includes the data collection process. The data points used in the training set $\mathcal{D}$ of ERM (4) must be collected in a way that is compliant with fundamental rights and regulations such as the General Data Protection Regulation [30], [115]. Data collection should be transparent and representative, avoiding biased sampling and improper consent procedures [116], [117].

B. Model

We can ensure the fairness of the learnt hypothesis $\hat{h} \in \mathcal{H}$ by using a model $\mathcal{H}$ that contains only hypothesis maps satisfying fairness constraints such as (variations) of (22). One example for such a constraint is to require each hypothesis h to be Lipschitz continuous [101]. The idea is to require $\hat{h}$ to deliver similar predictions for data points that are similar in a nondiscriminatory sense. A key challenge for the practical use of this requirement is to find a useful choice for the metric underlying the Lipschitz condition [101].

C. Loss

1) *Fairness Through Regularization*: Fairness constraints of the form (23) can be included in the loss of ERM. Using Lagrangian duality [37, Ch. 5], the constraints can be translated into a penalty term which is added to the ERM objective function [105], [107], [118], [119]. Adding such a fairness penalty term can be interpreted as a form of regularization (see Section II and Fig. 4).

2) *Fairness via Sample Weighting*: Instead of adding a penalty term to the loss function in ERM, we can also ensure fairness through sample weighting [120]. The rationale is to scale the loss incurred on a data point based on the relative frequency of its sensitive attribute in the training set $\mathcal{D}$. Magnifying the loss incurred for data points from a minority ensures that underrepresented groups have a greater influence on the solution of (4).

IX. KR6 - SOCIETAL AND ENVIRONMENTAL WELL-BEING

KR6 [20, p.19] covers the impact of AI systems on environmental and social well-being [121], [30, Article 35]. The EU Charter of Fundamental rights in Article 37 mandates a duty of the EU to provide for a high level of environmental protection and commitment to sustainable development [30]. This requirement therefore guides AI development to take environmental factors into account and thereby supports individual rights and collective welfare.

So far, we have discussed KRs that focus on the impact of ERM-based methods on individual users. In contrast, KR6 revolves around the broader impact of an ERM-based method at the societal and natural environment level.

A. Society and Democracy

Design choices for ERM should also consider the impact of (predictions delivered by) a trained model $\hat{h} \in \mathcal{H}$ on society as a whole. The predictions $\hat{h}(\mathbf{x})$ could not only be detrimental

to the mental health of individual users but also affect core democratic processes such as policy-making or elections. For example, social media apps train personalized models $\hat{h}$ to recommend (or select) content delivered to its users. The resulting tailored filtering of content can boost polarisation and, in the extreme case, social unrest [122].

B. Environment

ERM-based AI systems need to solve the optimisation problem (3) using some computational methods. Implementing these methods in physical hardware requires energy, which is typically provided in the form of electricity [123]. Given the increasing energy demand of AI systems, it is crucial to use environmentally friendly means of energy production [124]. Design choices for ERM should minimize the energy and cooling water requirements [125], of the resulting AI system. These requirements depend not only on the computational effort required to solve ERM (4) but also on the choice of data collection strategies [126].

X. KR7 - ACCOUNTABILITY

KR7 [20, p.19] supplements the other key requirements and is closely related to the key requirement of fairness (KR5). Accountability requires mechanisms to identify, explain, and address the potential harms of AI systems. Organizations operating an AI system are responsible for its direct and indirect effects on the user [127], [128]. Developers and deployers must take steps to explain the objectives, motivations, and reasons for the behaviour and outcome of AI systems. Accountability includes the reporting of data breaches and the possibility of redress [129].

A. Policy and Governance Approaches

Organizations such as the OECD have been working on governance structures for ERM-based AI systems. This includes formalising auditing procedures and ensuring that developers adhere to both ethical standards and legal requirements [130].

B. Frameworks for Answerability

AI developers and operators need to be able to justify their actions and decisions. This includes both transparency and oversight mechanisms, which are particularly important in high-stakes domains [131]. The justification of ERM design choices also requires a solid understanding of the inherent trade-offs between design criteria such as explainability and accuracy [95], [132].

C. Regular Audits and Third-Party Reviews

Periodic reviews of AI systems by independent auditors help ensure and validate accountability. To this end, independent external teams (“red teams”) should stress-test the ERM-based system for vulnerabilities and biases that could undermine accountability [130], [133].

Overall, accountability requires a good understanding of the entire design process of the AI system, as well as proper documentation and evaluation of the trade-offs during the design process and therefore can only be achieved if the other key requirements are taken seriously and well documented.

XI. CONCLUSION AND FUTURE WORK

It is essential for the future development of AI systems that research focuses on the creating trustworthy AI to ensure that any development is beneficial to individuals and society at large and is consistent with sustainability goals and democratic processes. AI systems pose an imminent risk of being harmful, biased, and opaque. To prevent negative impacts from the use of AI systems, it is also necessary to understand the core values we seek to protect to safeguard individuals and societies. Therefore, we structured our work by first providing an overview over core principles of trustworthy AI and their basis in European fundamental rights and second proposing different design choices that would allow us to realise those principles.

We proposed different design choices using ERM methods to realise the requirements of human agency and oversight, technical robustness and safety, privacy and data governance, transparency, diversity, nondiscrimination and fairness, societal and environmental well-being, and accountability.

Future work is needed to better understand potential trade-offs between these requirements and possible ways to mitigate their impact. While our contribution is to explore and propose different ways to address the key requirements proposed by the EU, more research is needed to gain a clearer understanding of specific use cases and potential harms that can be mitigated through design choices in the process of planning and building AI systems. Finally, more interdisciplinary research will be needed to examine the implementation of AI legislation and its limits.

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